Eigenvalues, Diagonalisation & Spectral Theory

Summary

Theory:

This file covers Matrix Theory and matrix decomposition

Content:

Eigenvalues, Eigenvectors, Similar Matrices,
Eigendecomposition & Spectral Decomposition

Eigenvalues & Eigenvectors

Definitions & The Core Equation

Eigenvector (x)

- A non-zero vector x such that when a given square matrix A is applied to it, the resulting vector is a scalar multiple of x (like invariant lines)

Eigenvalues (λ)

- The scalar λ associated with an eigenvector x such that the equation $Ax = \lambda x$ holds true. It tells you **how much** the eigenvector gets stretched or squashed (e.g. if $\lambda = 2$, the vector doubles in length)

The Core Equation

The relationship between a matrix A, its eigenvector x, and its eigenvalue λ is defined by:

$$Ax = \lambda x$$

Algorithm: Finding Eigenvalues & Eigenvectors

To find eigenvalues and eigenvectors we use the characteristic equation:

1. **Set up the Characteristic Equation**: Rearrange $Ax = \lambda x$ to $(A - \lambda I)x = 0$ (where I is the identity matrix). For x to be a non-zero vector, the matrix $(A - \lambda I)$ must not be invertible. Therefore, its determinant must be zero:

$$\det(A - \lambda I) = 0$$

2. **Solve for the Eigenvalues (λ)**: Compute the determinant of $(A - \lambda I)$. This will give you a polynomial equation (the characteristic polynomial). The eigenvalues are the roots of the polynomial
3. **Substitute λ back in**: Take your first eigenvalue and substitute it into the matrix $(A - \lambda I)$
4. **Solve for the Eigenvector (x)**: Solve the resulting homogeneous linear system $(A - \lambda I)x = 0$ (usually via Gaussian elimination) to find the components of the vector x
5. **Repeat**: Repeat steps 3 and 4 for every eigenvalue you found to get all corresponding eigenvectors

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Example: Finding Eigenvalues & Eigenvectors

Question: Find the eigenvalues and eigenvectors for the matrix A:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Step 1: Set up the Characteristic Equation

We need to find the determinant of $(A - \lambda I)$ and set it to zero. First, we subtract λ from the main diagonal of A:

$$A - \lambda I = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$$

Now, we calculate the determinant of this new matrix:

$$\det(A - \lambda I) = (4-\lambda)(3-\lambda) - (1)(2) = 0$$

Step 2: Solve for the Eigenvalues (λ)

Expand the brackets and simplify to find the characteristic polynomial:

$$\begin{aligned} (12 - 4\lambda - 3\lambda + \lambda^2) - 2 &= 0 \\ \lambda^2 - 7\lambda + 10 &= 0 \end{aligned}$$

Factorise the quadratic equation to find the roots (our eigenvalues):

$$(\lambda - 5)(\lambda - 2) = 0$$

So, our two eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$

Step 3 & 4: Substitute $\lambda_1 = 5$ and Solve for its Eigenvector

Substitute $\lambda = 5$ back into our $(A - \lambda I)$ matrix:

$$\begin{pmatrix} 4-5 & 1 \\ 2 & 3-5 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}$$

Set up the homogeneous system $(A - \lambda I)x = 0$:

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This gives us two equations, but they are multiples of each other (which is exactly what should happen!):

$$-x_1 + x_2 = 0 \implies x_1 = x_2$$

To find an eigenvector, we just pick a simple, non-zero value that satisfies this rule. Let $x_1 = 1$. Then $x_2 = 1$

Eigenvector 1:

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 5: Repeat for $\lambda_2 = 2$

Substitute $\lambda = 2$ back into our $(A - \lambda I)$ matrix:

$$\begin{pmatrix} 4-2 & 1 \\ 2 & 3-2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$$

Set up the homogeneous system:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This gives us the equation:

$$2x_1 + x_2 = 0 \implies x_2 = -2x_1$$

Again, pick a simple value. Let $x_1 = 1$. Then $x_2 = -2$.

Eigenvector 2:

$$v_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Eigenspaces & Multiplicity

Definitions

Eigenspace **(E_λ)

- The set of all eigenvectors corresponding to a specific eigenvalue λ_0, together with the zero vector. It is the null space (or kernel) of the matrix $(A - \lambda I)$
- If an eigenvalue is the "stretch factor", the eigenspace is the entire line, plane, or 3D space of all possible vectors that get stretched by that exact amount

Basis for an Eigenspace

- A set of linearly independent eigenvectors that spans the eigenspace E_λ

Algebraic Multiplicity

- The number of times a specific eigenvalue λ appears as a repeated root of the characteristic polynomial

Geometric Multiplicity

- The dimension of the eigenspace E_λ. This is equal to the number of linearly independent basis vectors that span E_λ

Example: Finding the Eigenspace & Multiplicity

Note that this is the exact same matrix we just solved:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Step 1: Check Algebraic Multiplicity from the Polynomial

Our characteristic polynomial was:

$$(\lambda - 5)(\lambda - 2) = 0$$

- The root $\lambda = 5$ appears exactly once. Therefore, its **Algebraic Multiplicity is 1**
- The root $\lambda = 2$ appears exactly once. Therefore, its **Algebraic Multiplicity is 1**

Step 2: Define the Eigenspace and Basis for $\lambda = 5$

When we solved $(A - 5I)x = 0$, we found the defining relationship was $x_1 = x_2$

The Eigenspace (E_5):

$\therefore$ This is the set of all vectors where the top and bottom numbers are identical

$$E_5 = \left\{ \begin{pmatrix} c \\ c \end{pmatrix} \middle| c \in \mathbb{R} \right\}$$

- **The Basis**: We factor out the constant c to find our core direction vector

$$\text{Basis for } E_5 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

- **Geometric Multiplicity**: Because our basis contains exactly 1 vector, the **Geometric Multiplicity is 1**

Step 3: Define the Eigenspace and Basis for $\lambda = 2$

When we solved $(A - 2I)x = 0$, we found the defining relationship was $x_2 = -2x_1$

- **The Eigenspace (E_2)**: This is the set of all vectors where the bottom number is negative two times the top number

$$E_2 = \left\{ \begin{pmatrix} c \\ -2c \end{pmatrix} \middle| c \in \mathbb{R} \right\}$$

- **The Basis**: We factor out the constant c to isolate the core vector.

$$\text{Basis for } E_2 = \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$$

- **Geometric Multiplicity**: Because our basis contains exactly 1 vector, the **Geometric Multiplicity is 1**

Similar Matrices & Eigendecomposition (Not Yet Covered)

Matrix Similarity

Conditions for Diagonalisation

The Eigendecomposition Formula

Spectral Decomposition (Not Yet Covered)

Symmetric Matrices

The Spectral Theorem

#linear-algebra/theory #linear-algebra/eigenvalues #linear-algebra/diagonalisation #linear-algebra/spectral-theory